



## Catizen: Building lag-free, accessible Web3 games with generative AI and Google Cloud

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### GOOGLE CLOUD RESULTS

- ✓ Enables smooth gaming experiences for 150,000 peak concurrent online users
- ✓ Minimizes blockchain transaction processing time to 2-3 seconds
- ✓ Powers speedy data reading and writing with Cloud SQL and Cloud Storage
- ✓ Generates more than 10 engaging videos per day with the Gemini API
- ✓ Supports 20 million user growth within three months

By deploying its Web3 game in Google Cloud, Catizen ensures quality gaming

# experiences for its fast-growing user base and leverages generative AI technology to quickly produce engaging marketing content.

Web3 games have revolutionized the gaming industry since 2023 by unlocking truly player-centric experiences. Leveraging the decentralized nature of blockchain technology, these new games give users more authority over their in-game items, allowing them to own and use them across different games. But despite the groundbreaking features of Web3 games, many Web2 gamers find the complex technology daunting. That's why [Catizen](#) exists: to make Web3 products more accessible.

"We firmly believe that Web3 technology can make gaming fairer and more rewarding for game developers and players," says Ricky Wong, Co-Founder & CFO at Catizen. "That's why, besides game development, we've been putting a lot of effort into promoting Web3. For example, by playing Catizen games, users can easily grasp Web3 concepts like staking and airdrops."

**"We needed a cloud data warehouse that supports high-performance data analytics, so that we could provide compelling content and realize accurate ad targeting based on data insights. Google Cloud provides the most cost-effective data analytics solution on the market."**

**Ricky Wong**  
Co-Founder & CFO,  
Catizen

This focus on offering excellent gaming experiences drives Catizen's game development. Founded in March 2024 with its flagship game Catizen, the team anticipated quick user growth. Since network speed is crucial for a real-time Web3 game that involves constant on-chain interactions, Catizen looked for a highly

scalable cloud infrastructure with fast connectivity to deploy its game. After comparing different offerings, it landed on Google Cloud for its outstanding scalability and network capabilities.



## Ensuring glitchless gaming experiences during exponential user growth

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In the early days of its relationship with Google Cloud, the Catizen team didn’t have an in-depth understanding of the various application scenarios that platform offers. But with the Google Cloud team’s proactive technical support and advice, Catizen quickly improved its architecture through resource and cost optimization.

“The experience of working with the Google Cloud team has been outstanding. They provided valuable insights on how we could maximize performance while minimizing costs, and made sure everything was configured correctly and running smoothly,” says Wong.

Catizen uses the virtual machine (VM) instances in Compute Engine as servers of its flagship game. Since the VMs automatically scale up based on user demand, this solution was essential to the success of the game, which gained 15 million players within 10 weeks. This high scalability ensured a glitchless gaming experience

when the number of players grew exponentially—even when concurrent online users hit 150,000 during a marketing campaign.

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## Enabling smooth and speedy Web3 gaming

One of the biggest challenges for Web3 game developers is to overcome the speed limitations imposed by decentralized transaction verification on blockchains. With the low latency of Google Cloud networking products like [Cloud Load Balancing](#), Catizen has optimized the payment processing speed of its game and can now complete a blockchain-based payment within two to three seconds, which is among the fastest in the Web3 gaming industry.

Furthermore, employing [Cloud SQL](#) and [Cloud Storage](#) to store its in-game data allows Catizen to make its Web3 game run as smoothly as a PC game. These solutions support quick data reading and writing speed, giving Catizen’s users instant responses whenever they click on something in the game.

“Being able to deliver smooth gaming experiences is important to us because speed is a common performance bottleneck for Web3 games,” explains Wong. “We’re able to achieve this thanks to the excellent network and database read and write capabilities of Google Cloud.”



## Enhancing user engagement and interaction with AI

Catizen’s popularity shows not only in its number of players but in the size of its online community. As of June 2024, the game had

more than four million channel subscribers on Telegram, an encrypted messaging app popular among crypto enthusiasts, and 1.6 million followers on X. David Mak, CEO at Catizen, notes that the [Gemini API](#), which Catizen's marketing team uses to generate at least 10 short videos per day, contributes significantly to the company's marketing success.

"Using the Gemini API, we can quickly produce appealing content for both entertainment and educational purposes, which has helped us build an engaging community in a short period of time," David Mak says. "It also allows us to instantly catch up with the latest trending topics on the internet and create videos that our users like."

Catizen also embraces other artificial intelligence (AI) tools from Google Cloud to enhance interactions among users and its customer service. The team uses [Translation AI](#) to realize seamless communication across languages for its global users and [Vertex AI](#) to power its gaming bots.



## Expanding Web3 services with Google Cloud

Following the impressive initial success of its first game, Catizen will expand its Web3 services in the coming months. It plans to add 18 Web3 games to its portfolio and evolve into a comprehensive mini-game platform, which will provide more AI-driven features using the Gemini API. The startup is also developing Web3-based ecommerce services and a short series application that will employ [Cloud CDN and Media CDN](#) for content distribution.

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**Established** in March 2024, **Catizen** is a Web3 mini-game developer and distributor dedicated to making Web3 products more accessible. Catizen, its flagship game launched on Telegram, currently has more than 20 million users across the world. Aiming at building a Web3 ecosystem, Catizen is expanding its game catalog and will provide mini-series streaming and e-commerce services in the near future.

**Industry:** Gaming

**Location:** Singapore

**Products:** [Google Cloud](#), [Cloud Load Balancing](#), [Cloud SQL](#), [Cloud Storage](#), [Compute Engine](#), [Gemini API](#), [Translation AI](#), [Vertex AI](#)

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